

CHAPTER 27

ELECTRICAL

SECTION BC 2701 GENERAL

2701.1 Scope. This chapter governs the electrical components, equipment and systems used in buildings and structures covered by this code. Electrical components, equipment and systems shall be designed and constructed in accordance with the provisions of the *New York City Electrical Code*.

SECTION BC 2702 EMERGENCY AND STANDBY POWER SYSTEMS

2702.1 Installation. Emergency power systems and standby power systems shall be installed in accordance with the *New York City Electrical Code*, NFPA 110 and NFPA 111.

2702.1.1 Fuel supply. Systems relying on fuel supplies shall have an on-premises fuel supply sufficient for not less than 6-hour full-demand operation of the system. However, natural gas from the public utility street main shall be permitted as the sole fuel supply for (i) emergency power systems serving Group R-2 occupancies, (ii) emergency power systems where permitted by Appendix G of this code, and (iii) standby power systems, provided that an outside gas cut-off valve separate from other gas services is installed in accordance with Section E.6 of Appendix E of the *New York City Fuel Gas Code*.

2702.1.2 Stationary generators. Stationary emergency and standby power generators required by this code shall be listed in accordance with UL 2200.

2702.1.2.1 Prohibited location. Emergency and standby power generators shall not be located in the same room as the main or primary electrical service equipment. Distribution associated with emergency and standby power systems shall not pass through the room containing the main or primary electrical service equipment.

2702.1.2.2 Enclosure. Emergency and standby generators within a building shall be located within a dedicated room separated by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 712, or both.

2702.1.2.2.1 Generator room contents. Equipment and fuel tanks located within the emergency generator room shall be limited to equipment and fuel tanks associated with the emergency and standby power systems. Piping within the room shall be limited to sprinkler piping.

Exceptions:

1. Uninterrupted conduits not associated with the emergency and standby power system may pass through the emergency generator room.
2. Piping associated with generator cooling systems may be located within the emergency generator room.

2702.1.2.3 Multiple generators. Multiple generators supplying emergency power system loads only, or supplying emergency power system equipment in combination with optional standby power loads as a common system may be located in the same room. Such generators may also share a common room, fuel supply and other common equipment and systems.

2702.1.3 Capacity. The emergency and standby power systems shall have capacity and ratings that supply all equipment required to be operational at the same time.

2702.1.4 Protection of fire pumps. Where electrically powered fire pumps are connected to the emergency power system, overcurrent protection and feeder conductors shall be provided in accordance with the *New York City*

Electrical Code. Automatic transfer switches shall be located in the same room as the fire pump as an integral part of the pump controller.

2702.1.5 Special inspection. The installation of a required emergency or standby power system shall be subject to special inspection in accordance with Section 1705.

2702.1.6 Registration. Fuel-burning equipment for an emergency or standby power system shall be registered with the Department of Environmental Protection in accordance with the requirements of Section 24-109 of the *Administrative Code*.

2702.1.7 Automatic transfer devices and power system feeders. Automatic transfer devices, emergency generators and emergency or standby power system feeders shall comply with Sections 2702.1.7.1 through 2702.1.7.2.1.

2702.1.7.1 Prohibited location. All automatic transfer devices and power system feeders that serve emergency and required standby power system equipment shall not be located in the same room as the emergency power system equipment or the main or primary electrical service equipment. Where emergency and standby transfer devices are installed in accordance with Articles 700 and 701 of the *New York City Electrical Code*, automatic transfer devices for optional standby power systems shall not be installed in the same room.

2702.1.7.2 Enclosure. Any automatic transfer device that is not located at the load shall be located within a dedicated room separated by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 712, or both.

2702.1.7.2.1 Automatic transfer device room. Equipment located within the automatic transfer device room shall be limited to equipment associated with the emergency or standby power systems. Piping within the room shall be limited to sprinkler piping.

2702.2 Where required. Emergency and standby power systems shall be provided where required by this section.

2702.2.1 Group A occupancies. Emergency power shall be provided for voice/alarm communication systems in Group A occupancies in accordance with Section 907.5.2.2.5.

2702.2.1.1 Prior code buildings. In prior code buildings, where a stationary generator is not otherwise required, the power source for emergency power to the voice/alarm communication system may be served by a gas generator or an uninterruptable power source (UPS) in accordance with the *New York City Electrical Code*.

2702.2.2 Smoke control systems. Standby power shall be provided for smoke control systems in accordance with Section 909.11.

2702.2.3 Exit signs. Emergency power shall be provided for exit signs in accordance with Section 1013.6.3.

2702.2.4 Means of egress illumination. Emergency power shall be provided for means of egress illumination in accordance with Section 1008.3.

2702.2.5 Accessible means of egress elevators. Standby power shall be provided for elevators that are part of an accessible means of egress in accordance with Section 1009.4.

2702.2.6 Horizontal sliding doors. Standby power shall be provided for horizontal sliding doors in accordance with Section 1010.1.4.3.

2702.2.7 Semiconductor fabrication facilities. Emergency power shall be provided for semiconductor fabrication facilities in accordance with Section 415.11.10.

2702.2.8 Membrane structures. Standby power shall be provided for auxiliary inflation systems in accordance with Section 3102.8.2. Emergency power shall be provided for exit signs in tents and membrane structures.

2702.2.9 Hazardous materials. Emergency or standby power shall be provided in occupancies with hazardous materials in accordance with the *New York City Fire Code* and Section 414.5.2 of this code.

2702.2.10 Highly toxic and toxic materials. Emergency power shall be provided for occupancies with highly toxic or toxic materials in accordance with the *New York City Fire Code*.

2702.2.11 Organic peroxides. Standby power shall be provided for occupancies with organic peroxides in accordance with the *New York City Fire Code*.

2702.2.12 Pyrophoric materials. Emergency power shall be provided for occupancies with silane gas in accordance with the *New York City Fire Code*.

2702.2.13 Covered and open mall buildings. Emergency power shall be provided for voice/alarm communication systems in covered and open mall buildings in accordance with Section 402.7.4.

2702.2.14 High-rise buildings. Emergency and standby power shall be provided in high-rise buildings in accordance with Section 403.4.8.

2702.2.15 Underground buildings. Emergency and standby power shall be provided in underground buildings in accordance with Section 405.8.

2702.2.16 Group I-3 occupancies. Emergency power shall be provided for doors in Group I-3 occupancies in accordance with Section 408.4.2.

2702.2.17 Elevators. Standby power for elevators, including elevators provided to accommodate ambulance stretchers pursuant to Section 3002.4, shall be provided as set forth in Section 3003.1.

2702.2.18 Airport traffic control towers. Standby power shall be provided in airport traffic control towers in accordance with Section 412.3.9.

2702.2.19 Smokeproof enclosures and pressurized elevator shafts. Standby power shall be provided for smokeproof enclosures as required by Section 909.20.6.2 and for pressurized elevator shafts provided in accordance with Section 3006.1.

2702.2.20 Occupancy Groups B, E and R-1. Emergency and standby power shall be required throughout those Group B, E and R-1 occupancies specified in this section:

1. Group B occupancies with occupied floor less than 75 feet (22 860 mm) above the lowest level of fire department vehicle access having a gross floor area over 15,000 square feet (1393.6 m²) on each floor, or having a total gross floor area of 100,000 square feet (9290.3 m²) or more.
2. Group E occupancies with occupied floor less than 75 feet (22 860 mm) above the lowest level of fire department vehicle access having a gross floor area over 15,000 square feet (1393.6 m²) on each floor, or having a total gross floor area of 100,000 square feet (9290.3 m²) or more.
3. All Group R-1 occupancies.

2702.2.20.1 Equipment requiring emergency power system. With respect to such Occupancy Groups B, E and R-1, the following equipment, where such equipment is required by this code, shall be provided with an emergency power system:

1. Exit signs and means of egress illumination required by Chapter 10;
2. Elevator car lighting;
3. Emergency voice/alarm communications systems, including Fire Department in-building Auxiliary Radio Communication systems (ARCs) provided where required or installed voluntarily in accordance with Section 916;
4. Automatic fire detection systems;
5. Fire alarm systems; and
6. Electrically powered fire pumps, including manual fire pumps, automatic fire pumps and sprinkler booster pumps.

2702.2.20.2 Equipment requiring standby power system. With respect to such Occupancy Groups B, E and R-1, the following equipment, where such equipment is required by this code, shall be provided with a standby power system:

1. Ventilating systems used for smoke venting or control;
2. Stair pressurization;
3. At least three elevators in a building at one time with manual transfer to other elevators as required by Section 3003; and

ELECTRICAL

4. Power and lighting for the fire command center required by Section 403.4.6.

2702.2.21 Continuously operating HVAC systems. Standby power systems shall be provided for fans or fan systems being designed in accordance with Section 607.5 of the *New York City Mechanical Code* and Section 717 of this code.

2702.2.22 Gas detection systems. Emergency power systems shall be provided for gas detection systems in accordance with Section 918.5.

2702.3 Maintenance. Emergency and standby power systems shall be maintained and tested in accordance with the *New York City Fire Code* and *New York City Electrical Code*.

2702.4 Required loads for optional standby power systems. In addition to any other loads, optional standby power systems shall be capable of providing power to the following standby and emergency power loads upon failure of the normal power supply:

1. Emergency lighting;
2. Fire alarm systems; and
3. Elevators as follows:
 - 3.1. For Group R-2 occupancies in buildings greater than 125 feet (38 100 mm) in height, at least one elevator serving all floors, or one elevator per bank where different banks serve different portions of the building; or
 - 3.2. For all other buildings having occupied floors located more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access, at least one elevator that serves all floors.

Exception: Backup power systems that provide backup power for non-accessory public telecommunications equipment, such as rooftop telecommunications antennas, cooling equipment, routers, etc., need not provide power to the standby and emergency power loads listed in this section.

2702.4.1 Additional optional standby power systems. Where a functioning emergency power system or required standby power system is in place and provides power to all required emergency or required standby power loads, any additional optional standby power system need not supply power to such emergency or required standby power loads.

2702.4.2 Compliance with NFPA 110 and UL 2200. Where an optional standby power system is required to supply emergency or required standby power loads, such power system shall comply with the UL 2200 listing and NFPA 110.

2702.4.3 Special inspection. Where an optional standby power system is required to supply emergency or required standby power loads, such power system shall be subject to special inspection in accordance with Section 1704.31 and the rules of the department.